

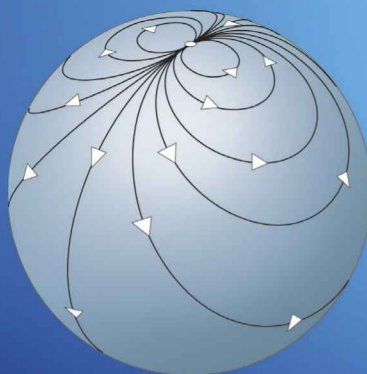
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VISUAL COMPLEX ANALYSIS

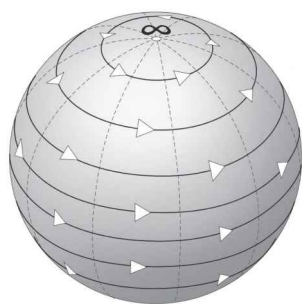
TRISTAN NEEDHAM

Foreword by
ROGER PENROSE



25TH ANNIVERSARY EDITION

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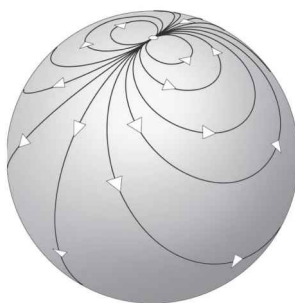


VISUAL COMPLEX ANALYSIS

TRISTAN NEEDHAM

UNIVERSITY OF SAN FRANCISCO

Foreword by
ROGER PENROSE



25TH ANNIVERSARY EDITION

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*For
Roger Penrose
and
George Burnett-Stuart*

FOREWORD

Roger Penrose

Complex analysis is the theory of functions of complex numbers or, more specifically, *holomorphic* functions of such numbers. This theory is both profoundly beautiful and vastly influential, both in pure mathematics and in many areas of application, particularly in physics, indeed being central to the underlying formalism of quantum mechanics. However, the very concept of a complex number is an essentially abstract one, depending upon the seemingly absurd notion of a square root of -1 , the square of any ordinary real number being, unlike -1 , necessarily non-negative.

Yet, it should be borne in mind that even the notion of a so-called “real” number is also an abstraction, and we must move far beyond the immediate notion of “counting numbers” $0, 1, 2, 3, 4, \dots$, and beyond even the fractions $\frac{1}{2}, \frac{3}{5}, -\frac{7}{4}$, etc., if we are to express even the square root of 2. But here we are helped by a *visual image*, and can perceive a straight line extended indefinitely in both directions to give us a good intuitive impression of the full array of real numbers. The slightly misleading term “real” for this imagined array is excusable, as we can indeed imagine a ruler, or a line of ink drawn on a piece of paper, as providing us with some sort of conceptual image of this array. This greatly helps our understandings of what the mathematician’s precise notion of a “real number” is intended to idealize. We are not concerned with whatever might be the nature of the physics of the particles or fields that might compose our ruler or ink-line; nor, indeed, do we require any concept of the cosmology that may be relevant to the extension out to infinity of our imagined ruler or ink-line. Our abstract mathematical notion of a “real number” remains aloof from any such realities of the actual world. Yet, in a curious reversal of roles, it is this very mathematical idealization that underlies most of our theories of the actual world.

So, what can be the driving force behind a need to go beyond these seemingly ubiquitous “real numbers”? What purpose might there be for the introduction of a “square root of -1 ”? Such a number fits nowhere within the span of the real numbers, and it would appear that we have no reason to demand that the equation $x^2 + 1 = 0$ have any kind of “solution”. The answer to this desire for such entities lies in the *magic* that lies hidden within them, but it is not a magic that immediately reveals itself. In fact, when the first hint of this magic was actually perceived, in

the mid 16th century, by Girolamo Cardano, and then more completely by Rafael Bombelli, these strange numbers were dismissed, even by them, as being as useless as they were mysterious.

It is of some interest to note that it was not in the equation $x^2 + 1 = 0$ that this hint of magic was first perceived, but in cubic equations like $x^3 = 15x + 4$, which has the perfectly sensible solution $x = 4$. Yet, as Bombelli had noted, Cardano's general expression for the solution of such cubic equations necessarily involves a detour into a mysterious world of numbers of a sort where the equation $x^2 + 1 = 0$ is deemed to have the two solutions, now referred to as the *imaginary units* $x = i$ and $x = -i$, of an algebra—now called *complex-number algebra*—that had appeared to be consistent, but not what had been regarded as “*real*”.

This dismissive attitude did not change much until the mid to late 18th century, with Leonhard Euler's remarkable formula $e^{i\theta} = \cos \theta + i \sin \theta$ and, even more importantly, the geometrical representation of the entire family of complex numbers as points in a Euclidean plane, as initially proposed by Caspar Wessel, where the algebraic operations on complex numbers are readily understood in geometrical terms. This provided a kind of 2-dimensional “visual reality” to the array of complex numbers that could be combined with topological notions, such as employed initially by Carl Friedrich Gauss, and soon followed by others. The early to mid 19th century saw many important advances, many of these being due to Augustin-Louis Cauchy, especially with the beauty and the power of contour integration, and, perhaps most profoundly, with ideas due to Bernhard Riemann. The very notion of “complex smoothness” of complex functions was expressed by use of the Cauchy–Riemann equations, and this provided the powerful concept of a *holomorphic function* that implies that a power-series expansion always locally exists, this leading to a vast and powerful theory with numerous magical properties.

The two revolutions of early 20th century physics both owe a profound debt to complex-number mathematics. This is most manifest with quantum mechanics, since the basic formalism of that theory depends fundamentally on complex numbers and holomorphic functions. We see a remarkable interplay between quantum spin and the geometry of complex numbers. The basic equations of Schrödinger and Dirac are both complex equations. In relativity theory, the transformations relating the visual field of two observers passing close by each other at different relativistic speeds is most easily understood in terms of simple holomorphic functions. Moreover, many solutions of Einstein's equations for general relativity benefit greatly from properties of holomorphic functions, as does the description of gravitational waves.

In view of the undoubted importance of complex analysis in so much of mathematics and physics, it is clearly important that there are basic accounts of these topics available to those unfamiliar (or only partly familiar) with the basic ideas

of complex analysis. In this foreword I have very much stressed how the visual or geometric viewpoint has been of vital importance, not only to the historical development of complex analysis, but also to the proper understanding of the subject. Tristan Needham's *Visual Complex Analysis* as originally published in 1997 was, to my knowledge, unique in the extent to which it was able to cover these fundamental ideas with such thoroughness, visual elegance, and clarity.

With this *25th Anniversary Edition* there have been some significant improvements, most particularly in the incorporation of captions to the diagrams. This makes it easier for the reader to dip into the arguments, as illustrated so elegantly in such wonderfully expressive pictures, without necessarily having to look through to find the relevant portion of the text. In any case, I am sure that readers, over a broad range of relevant knowledge—from those with no prior experience of complex analysis to those already experts—will gain greatly from the charm, distinct originality, and visual clarity of the arguments presented here.

PREFACE TO THE 25TH ANNIVERSARY EDITION

Introduction

Mathematical reality exists outside the confines of space and time, but books about mathematics do not. A quarter century after its publication, I am grateful that an entirely new generation of mathematicians and scientists has continued to embrace VCA's unorthodox, intuitive, and, above all, *geometrical* approach to Complex Analysis. To mark the occasion, Oxford University Press has graciously permitted me to revisit the work, resulting in the creation of this significantly improved *25th Anniversary Edition*.

Principal Changes in the New Edition

Perhaps I should begin by noting what has *not* changed. As I shall elaborate below, back in 1997, one of my key mathematical innovations was the application of Newton's geometrical methods from the *Principia* (now 335 years old) to the Complex Analysis of Cauchy and Riemann (now more than 200 years old). Therefore, I have not seen any need to update the main body of the text, which has remained almost unchanged, save for a few additional corrections.

All of the changes to this new edition of VCA—both in appearance and substance—came about as a direct result of the knowledge I gained creating my second and final book,¹ *Visual Differential Geometry and Forms: A Mathematical Drama in Five Acts*, Princeton University Press, 2021 (hereafter abbreviated to VDGF). In particular, I thank Wanda España of PUP for her beautiful work on the design of VDGF, for this in turn inspired some design elements that I requested of OUP for this new edition of VCA. Of course I also thank Oxford University Press for *accepting* some of these suggestions, while also imprinting their own distinctive style on the work.

- The most obvious change is in the physical dimensions of the book: it has expanded from 6"×9" to 7"×10". There are two significant advantages to this change: (1) the book is more comfortable to hold and read; (2) the 503²

¹ [Needham \(2021\)](#).

² The original edition only contained 501 figures: The figure in this Preface and [6.41] are both new.

figures—the beating heart of the work!—are now 36% larger, and, one may dare to hope, 36% clearer!

- Another obvious change is the introduction of the standard numbering system for sections, subsections, equations, and figures. In hindsight, my deliberate avoidance of the standard scheme was perhaps a rather childish, tantrum-like expression of my disgust with the prevailing, life-sucking reduction of wildly exciting mathematical ideas to the arid structure enshrined in traditional treatises: “Lemma 12.7.2 implies Theorem 14.3.8”. Let me take this opportunity to apologize—better 25 years late than never?—to every professor who was ever brave enough to adopt VCA as the text for their course, only to discover that they had to struggle to refer their students to any given figure or result!
- The *Bibliography* has been updated. Not only have previously cited works been updated to their latest editions, but I have also added a number of *new* works, which were not cited in the first edition, for the simple reason that 25 years ago they did not *exist*!
- The *Index* has been improved significantly. Many new entries have been added, and, perhaps more importantly, wherever a single main entry formerly listed a long, frustratingly unhelpful string of undifferentiated page numbers, I have now split it into many individual and *helpfully specific subentries*. By way of proof, consider the entry for “Jacobian matrix”, then and now!
- *Giving credit to Eugenio Beltrami*. In the original edition I pointed out that in 1868 Beltrami discovered³ (and published) the conformal models of the hyperbolic plane, which Poincaré then rediscovered 14 years later, in 1882. Now, following the example I set myself in VDGF, I have gone one step further, attempting to put the record straight by *renaming* these models as the *Beltrami–Poincaré* disc and half-plane. Beltrami also discovered the projective model, and I have renamed that, too, as the *Beltrami–Klein* model. Correspondingly, my fancifully named inhabitants of these hyperbolic worlds have been renamed from “Poincarites” to *Beltrami–Poincarites*!
- Twenty five years ago, VCA was on the bleeding edge of what was *typographically* possible. Indeed, my editor told me that VCA was the *first* mathematics book published by Oxford University Press to be composed in $\text{\LaTeX}$ and yet *not* to be typeset using the standard Computer Modern fonts designed by Donald Knuth, the creator of $\text{\TeX}$. I was able to achieve this feat by virtue of Michael Spivak’s⁴ then newly created *MathTime* typeface, and by virtue of the wonderful

³ See [Milnor \(1982\)](#), [Stillwell \(1996\)](#), and [Stillwell \(2010\)](#).

⁴ Yes, the famous differential geometer!

Y&Y $\text{\TeX}$ System for Windows.⁵ I was thereby able to typeset the book in Times text and (mainly) MathTime mathematics—a *vast* aesthetic improvement over Computer Modern, in my humble but strong opinion!

Needless to say, the $\text{\TeX}$ world has moved on considerably in the past 25 years. But in order to take advantage of these advances, I was forced to grapple with the task of updating my original $\text{\TeX}$ files (including my countless macros) from the ancient (dead?) language of $\text{\LaTeX}$ 2.09 to the modern $\text{\LaTeX}$ 2 ϵ . This effort was rewarded with the ability to typeset this new edition of VCA using the same typefaces that I very carefully selected for VDGF, all three of which sprang from the genius mind of Hermann Zapf (1918–2015): Optima for the headings, Palatino for the text, and, crucially, the remarkable Euler fonts for the *mathematics*.

Let me pause for a moment to pay tribute to Zapf’s Euler mathematical fonts. They would appear to me to be at home onboard the starship of an advanced alien civilization, yet they also evoke the time-worn stone engravings within an ancient Greek ruin—their beauty transcends space and time: Picture an inscription on the *Guardian of Forever*!⁶

But my fatal attraction to these fonts brought about conflict between the Euler mathematics in the text of the new edition and the *MathTime* mathematics in all the *figures* of the original edition. Having been born cursed (and I suppose blessed) with a compulsion to strive for perfection, I had no choice but to undertake the self-inflicted, Herculean task of hand-editing (within CorelDRAW) all 501 of my original, hand-drawn figures! I then output new versions in which each MathTime symbol is here replaced with its matching Euler counterpart, thereby bringing the figures into perfect alignment with the new text.

I am the impossibly proud father of remarkable twin daughters, Faith and Hope. I am also the father of VCA and of VDGF. It is therefore a source of deep, resonating joy that *both* sets of twins now *look* like twins!

- Now let me explain the most fundamental change, the one that took me the greatest time and effort to accomplish, and the one that I believe transforms this *25th Anniversary Edition* into a truly *new* edition, one that may even be of value to owners of the original edition.

The 503 figures are the mathematical *soul* of the work. They crystallize all the geometrical insights I was able to glean from my many years spent struggling to *understand* Complex Analysis. Yet, in the entire original edition, *you will not find a single caption*. Why?!

Sadly, the answer is simple: *cowardice*. As a newly minted DPhil student of Penrose, with no track record or reputation, I feared that the mathematical community

⁵ Sadly, this system (designed by Professor Berthold Horn of MIT) ceased to exist in 2004.

⁶ The space-time portal featured in the *Star Trek* episode, “The City on the Edge of Forever”.

would reject or even ridicule⁷ my Newtonian arguments. I therefore sought to control the narrative (literally!) by only revealing the arguments within the text proper, where I could spell out my reasoning in full.

Well, 25 years later, I am certainly older, perhaps a tad wiser, and I have also gained a modicum of confidence by virtue of the enthusiastic reviews that VCA has since received in all the major journals. Perhaps more emotionally significant to me has been the large number of *individual* notes of appreciation I have received, which I continue to receive to this day, from readers of all stripes—graduate students, professors, and working scientists—from around the globe.

So, finally, I have done something that I should have had the courage to do at the outset:

In this 25th Anniversary Edition, every figure now has a caption that fully explains its mathematical content. Additionally, many of these new captions include a title (in bold print) that further distills the figure's meaning down to its essence.

This approach⁸ is directly inspired by the works of my teacher, friend, and mentor, Sir Roger Penrose—to whom this book is dedicated, and to whom I now offer sincere thanks for his generous Foreword!—most remarkably in his *Road to Reality*,⁹ where a single figure's caption can take up a quarter of a page!

This innovation now makes it possible to read VCA in an entirely new way—as a highbrow comic book! *Much* of the geometrical reasoning of the work can now be grasped simply by studying a figure and its accompanying caption, only turning to the main text for the complete explanation as needed. Furthermore, instead of undertaking a systematic, linear reading of the work, you are now invited to skip and hop about, lighting upon whichever figure happens to catch your eye.

- Newton's concept of *ultimate equality* underlies many of the arguments in this book, but its use was not made *explicit* in the original edition, and this led some to suppose that the arguments were less rigorous than they actually were (and remain). In this new edition I have only occasionally modified the main text, but I *have* felt at liberty to explicitly introduce ultimate equalities into some of the newly minted *captions*. As in my earlier works,¹⁰ I have used the symbol $\asymp$ to denote this concept.

Let me now expand upon this vitally important point . . .

⁷ In my defence, such fears were not entirely groundless: When Princeton University Press sent out draft chapters of VDGF for review, one of the three anonymous reviewers bluntly declared, "This is not even mathematics!"

⁸ This is also my approach in VDGF.

⁹ Penrose (2005).

¹⁰ Needham (1993), Needham (2014).

More on Newton's *Principia* and His Concept of *Ultimate Equality*

As I explain in the original Preface (immediately following this new Preface), VCA (and VDGF) could never have come to be had I not undertaken a careful study of Newton's *Principia* in 1982, while I was still a DPhil student of Penrose. As I struggled to penetrate Newton's ancient diagrams, I intuitively sensed the power and beauty of his geometrical reasoning, long before I fully understood it.

But one aspect of Newton's thinking did make sense to me *immediately*: why should Newton settle for studying an *equation* describing the orbit of a planet around the Sun when he possessed a geometrical tool of enormous power, capable of instead analyzing the orbit *itself*?! This in turn provoked within me a life-changing *mathematical crisis of conscience*: how many other mathematical phenomena had I fooled myself into *pretending* I had understood, when all I had done was grasp an equation that *described* the phenomenon from afar, rather than daring to stare mathematical reality squarely and *geometrically* in the face?

I felt excited and strangely fearful, like one of the apes in the opening of Stanley Kubrick's, *2001: A Space Odyssey*, shrieking and wildly leaping about in the stark silent presence of the black monolith. I wanted to be that one ape that was brave enough to *touch* the monolith, now taking the form of a different ancient reservoir of deeply alien knowledge: Newton's mind!

So I did. And once I had done so, I stared down at Newton's geometrical concept of *ultimate equality*, as the ape had stared down at the large femur bone on the ground before him, picked it up, gingerly at first, examined it closely, and then spent the next 35 years of my life wielding it with all the force and ingenuity I could muster—a tool to crack open and lay bare to our visual intuition the secrets of Complex Analysis, and, 25 years later, the secrets of Differential Geometry and Forms.

The presentation of the geometrical reasoning in VDGF *appears* to be more rigorous than that in VCA because there I *explicitly* make use of Newton's concept of ultimate equality. Let me now steal an excerpt from the Prologue of VDGF in order to explain what this means.

As I have discussed elsewhere,¹¹ Newtonian scholars have painstakingly dismantled the pernicious myth¹² that the results in the 1687 *Principia* were first derived by Newton using his original 1665 version of the calculus, and only later recast into the geometrical form that we find in the finished work.

Instead, it is now understood that by the mid-1670s, having studied Apollonius, Pappus, and Huygens, in particular, the mature Newton became disenchanted with

¹¹ See Needham (1993), the original 1997 Preface to VCA, and Needham (2014).

¹² Sadly, this myth originated with Newton himself, in the heat of his bitter priority battle with Leibniz over the discovery of the calculus. See Arnol'd (1990), Bloye and Huggett (2011), de Gandt (1995), Guicciardini (1999), Newton (1687, p.123), and Westfall (1980).

the form in which he had originally discovered the calculus in his youth—which is different again from the Leibnizian form we all learn in college today—and had instead embraced purely geometrical methods.

Thus it came to pass that by the 1680s Newton’s algebraic infatuation with power series gave way to a new form of calculus—what he called the “synthetic method of fluxions”¹³—in which the geometry of the Ancients was transmogrified and reanimated by its application to shrinking geometrical figures in their moment of vanishing. *This* is the potent but non-algorithmic form of calculus that we find in full flower in his great *Principia* of 1687.

Let me now immediately spell out Newton’s approach, and in significantly greater detail than I did in the first edition of VCA, in the vain hope that this new edition may bolster my efforts in VDGF to inspire more mathematicians and physicists to adopt Newton’s intuitive (yet rigorous¹⁴) methods than did the first edition.

If two quantities A and B depend on a small quantity ϵ , and their ratio approaches unity as ϵ approaches zero, then I shall avoid the more cumbersome language of limits by following Newton’s precedent in the *Principia*, saying simply that, “ A is ultimately equal to B ”. Also, as I did in earlier works [(Needham, 1993), (Needham, 2014)], I shall employ the symbol $\asymp$ to denote this concept of *ultimate equality*.¹⁵ In short,

$$“A \text{ is ultimately equal to } B” \iff A \asymp B \iff \lim_{\epsilon \rightarrow 0} \frac{A}{B} = 1.$$

It follows [exercise] from the theorems on limits that ultimate equality is an equivalence relation, and that it also inherits additional properties of ordinary equality, e.g.,

$$X \asymp Y \ \& \ P \asymp Q \implies X \cdot P \asymp Y \cdot Q, \quad \text{and} \quad A \asymp B \cdot C \iff (A/B) \asymp C.$$

Before I begin to apply this idea in earnest, I also note that the jurisdiction of ultimate equality can be extended naturally to things other than numbers, enabling one to say, for example, that two triangles are “ultimately similar”, meaning that their angles are ultimately equal.

As I explain in the original Preface, having grasped Newton’s method, I immediately tried my own hand at using it to simplify my teaching of introductory calculus, only later realizing how I might apply it to Complex Analysis (in VCA), and later still to Differential Geometry (in VDGF). Though I might choose any number of simple, illustrative examples [see Needham (1993) for more], I will reuse the specific one I gave in the original Preface to VCA, and for one simple reason: *this*

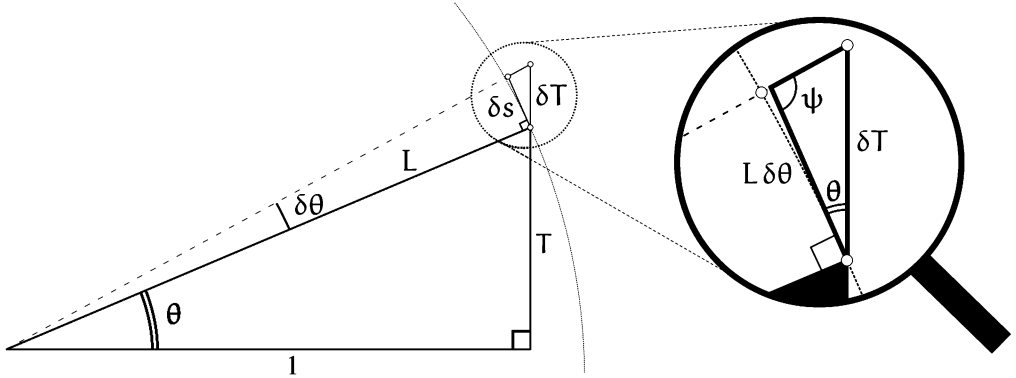
¹³ See Guicciardini (2009, Ch. 9).

¹⁴ Fine print to follow!

¹⁵ This notation was subsequently adopted (with attribution) by the Nobel physicist, Subrahmanyan Chandrasekhar (see Chandrasekhar, 1995, p. 44).

time I will use the “ $\asymp$ ”-notation to *present* the argument rigorously, whereas in the first edition I did not. Indeed, this example may be viewed as a recipe for transforming many of VCA’s “explanations” into “proofs”,¹⁶ merely by sprinkling on the requisite $\asymp$ ’s. With the addition of figure captions, I am now able to do some of this sprinkling myself, but some must still be left to the reader.

Before looking at this example, I suggest you first read the original presentation of the argument in the original Preface (immediately following this one) and *then* return to this point.



Let us show that if $T = \tan \theta$, then $\frac{dT}{d\theta} = 1 + T^2$ (see figure). If we increase θ by a small (ultimately vanishing) amount $\delta\theta$, then T will increase by the length of the vertical hypotenuse δT of the small triangle, in which the other two sides of this triangle have been constructed to lie in the directions $(\theta + \delta\theta)$ and $(\theta + \frac{\pi}{2})$, as illustrated. To obtain the result, we first observe that in the limit that $\delta\theta$ vanishes, the small triangle with hypotenuse δT is ultimately similar to the large triangle with hypotenuse L , because $\psi \asymp \frac{\pi}{2}$. Next, as we see in the magnifying glass, the side δs adjacent to θ in the small triangle is ultimately equal to the illustrated arc of the circle with radius L , so $\delta s \asymp L \delta\theta$; note that we have moved the dot from the corner of the triangle to the arc, to stress this point. Thus,

$$\frac{dT}{L d\theta} \asymp \frac{\delta T}{L \delta\theta} \asymp \frac{\delta T}{\delta s} \asymp \frac{L}{1} \implies \frac{dT}{d\theta} = L^2 = 1 + T^2.$$

So far as I know, Newton never wrote down this specific example, but compare the illuminating directness of his *style* of geometrical reasoning with the unilluminating computations we teach our students today, more than three centuries later! As Newton himself put it,¹⁷ the geometric method is to be preferred by virtue of the “clarity and brevity of the reasoning involved and because of the simplicity of the conclusions and the illustrations required.” Indeed, Newton went even further, resolving that *only* the synthetic method was “worthy of public utterance”.

¹⁶ I was already using the $\asymp$ notation (both privately and in print) at the time of writing VCA, and, in hindsight, it was a mistake that I did not employ it throughout the original edition of VCA.

¹⁷ See [Guicciardini \(2009, p. 231\)](#)

Newton himself did not employ *any* symbol to represent his concept of “ultimate equality”. Instead, his devotion to the geometrical *method* of the Ancients spilled over into emulating their *mode* of expression, causing him to write out the words “ultimately have the ratio of equality” every single time the concept was invoked in a proof. As [Newton \(1687, p. 124\)](#) explained, the *Principia* is “written in words at length in the manner of the Ancients”. Even when Newton claimed that two ratios were ultimately equal, he insisted on expressing *each ratio* in words. As a result, I myself was quite unable to follow Newton’s reasoning without first transcribing and summarizing each of his paragraphs into “modern” form (which was in fact already quite common in 1687). Indeed, back in 1982, this was the catalyst for my private introduction and use of the symbol $\asymp$.

It is my view that Newton’s choice *not* to introduce a symbol for “ultimate equality” was a tragically consequential error for the development of mathematics. As Leibniz’s symbolic calculus swept the world, Newton’s more penetrating geometrical method fell by the wayside. In the intervening centuries only a handful of people ever sought to repair this damage and revive Newton’s approach, the most notable and distinguished recent champion having been V. I. Arnol’d¹⁸ [1937–2010].

Had Newton shed the trappings of this ancient mode of exposition and instead employed some symbol (*any* symbol!) in place of the words “ultimately have the ratio of equality”, his dense, paragraph-length proofs in the *Principia* might have been reduced to a few succinct lines, and his mode of thought might still be widely employed today. Both VCA and VDGF are attempts to demonstrate, very concretely, the continuing relevance and vitality of Newton’s geometrical approach, in areas of mathematics whose discovery lay a century in the future at the time of his death in 1727.

Allow me to insert some fine print concerning my use of the words “rigour” and “proof”. Yes, my occasional explicit use of Newtonian ultimate equalities in this new edition represents a quantum jump in rigour, as compared to my original exposition in VCA, but there will be some mathematicians who will object (with justification!) that even this increase in rigour is insufficient, and that *none* of the “proofs” in this work are worthy of that title, including the one just given: I did not actually prove that the side of the triangle is ultimately equal to the arc of the circle.

I can offer no *logical* defence, but will merely repeat the words I wrote in the original Preface to VCA, 25 years ago: “... suppose one believes, as I do, that our mathematical theories are attempting to capture aspects of a robust Platonic world that is not of our making. I would then contend that an initial lack of rigour is a small price to pay if it allows the reader to see into this world more directly and

¹⁸ See, for example, [Arnol’d \(1990\)](#).

pleasurably than would otherwise be possible.” So, to preemptively address my critics, let me therefore concede, from the outset, that when I claim that an assertion is “proved”, it may be read as, “*proved beyond a reasonable doubt*”!¹⁹

The Continued Relevance of the Unorthodox *Contents* of VCA

Leaving aside its unorthodox geometrical *methods*, VCA is also distinguished by its unorthodox *contents*. The concept of the complex derivative (or *amplitwist*, as I call it) is not even introduced until Chapter 4, and many of the most interesting, unusual, and important parts of the book make little or no use of analysis.

For many decades before VCA, and now for decades after VCA, almost all²⁰ introductions to Complex Analysis have seemed to follow almost exactly in the footsteps of the ones that came before—the same topics, explained in the same order, and in the same manner, like a single-file procession of monks through the snow, quietly and sombrely intent on stepping into the footprints of the monk who went before them: Do not disturb the pristine snow lying all about you! But when I embarked upon VCA, 35 years ago, I was like a happily disobedient child, jumping about wildly, rolling my whole mind in the snow!

I started from scratch, asking myself, *which ideas connected to Complex Analysis have become the most vital to modern mathematics and physics?* A pair of closely related answers to this question immediately presented themselves, and while I make no claim to prescience, it is certainly true that the following have only become *more* vital to mathematics and physics over the past 25 years:

- **Hyperbolic Geometry** violates the normal rules of Euclidean geometry, and it is therefore also called *Non-Euclidean Geometry*. In the hyperbolic plane there are infinitely many different lines through a given point that are parallel to a given line, and the angles in a triangle always add up to *less* than π , the departure from π being proportional to the *area* of the triangle.

Yet, as Poincaré was the first to recognize, this strange geometry arises naturally across many parts of mathematics and physics. For example, the final figure of this book reveals how hyperbolic geometry unifies *all* the methods of solving the two-dimensional Dirichlet Problem. Furthermore, the visionary insights of Thurston²¹

¹⁹ Upon reading these words, a strongly supportive member of the Editorial Board of Princeton University Press suggested to my editor that in place of “Q.E.D.”, I conclude each of my proofs in VDGF with the letters, “P.B.R.D.”!

²⁰ I freely admit that I have not undertaken a systematic study of all the Complex Analysis textbooks that have been published during the last 25 years, and I am aware that there exist excellent *exceptions* to the following generalization, my favourite ones being [Shaw \(2006\)](#) and [Stewart and Tall \(2018\)](#), which I highly recommend precisely *because* of their very unusual contents. Incidentally, Shaw just happens to have been a fellow student of Penrose!

²¹ For details of Thurston’s *Geometrization Conjecture*—now *Theorem*!—see [Thurston \(1997\)](#).

(1946–2012)—subsequently vindicated by Perelman in 2003, six years after the publication of VCA—have established that hyperbolic geometry is in some sense more *fundamental* than Euclidean geometry.

This fascinating and important geometry is intimately entwined with the *complex numbers*, via the Möbius transformations that we shall discuss next. Yet, after a quarter century, VCA's long Chapter 6 remains the most complete and the most *geometrical* treatment of hyperbolic geometry that I have seen in any introduction to Complex Analysis.

- **Möbius Transformations** are mappings of the form

$$z \mapsto M(z) = \frac{az + b}{cz + d},$$

where a, b, c, d are complex constants. Whereas hyperbolic geometry is rarely even mentioned in introductory texts on Complex Analysis, these transformations are discussed in *all* of them, but usually in a perfunctory and superficial manner. In stark contrast, my long Chapter 3 remains the most complete and the most *geometrical* treatment of these transformations that I have seen in any introductory text.

The reason I lavished such extravagant attention upon these deceptively simple transformations is that they are possessed of *magical* powers, manifesting themselves in multifaceted guises across mathematics and physics. Again, it was Poincaré who was the first to recognize this. They are the isometries of both the *two-dimensional* hyperbolic plane (introduced above) and of *three-dimensional* hyperbolic space; they are the famous *Lorentz transformations* (isometries) of Minkowski and Einstein's *four-dimensional* spacetime; and when the constants are all integers, and $ad - bc = 1$, they form the *modular group*, describing the symmetries of the *modular functions* so important in modern number theory; and the list goes on . . .

The second pair of innovations with respect to the contents of VCA is centred on the use of *vector fields* as an alternative means of visualizing complex mappings. Instead of picturing a point z in $\mathbb{C}$ as being mapped to another point $w = f(z)$ in another copy of $\mathbb{C}$ —which is the paradigm in force throughout the first nine chapters—we instead *picture* $f(z)$ as a *vector emanating from* z .

- **The Topology of Vector Fields** is the subject of Chapter 10. Here we shed new light on the vital topological concept of the *winding number* (the subject of Chapter 7) by instead viewing it through the prism of the *index* of a singular point of a vector field. The climax of the topological analysis is a proof of the glorious Poincaré–Hopf Theorem, (10.4), which relates the indices of a flow on a closed surface to the surface's topological *genus*, which counts how many holes the surface has.

So far as I can tell, my final²² major innovative topic in VCA has *still* not had the impact I believe it deserves, despite my best efforts, and despite the even earlier, independent efforts of its first principal champion, Professor Bart Braden, who should also be credited²³ with having *named* the concept in honour of its first proponent:

- **The Pólya Vector Field** of a complex mapping $z \mapsto f(z)$ was introduced by George Pólya (1887–1985) in his 1974 textbook with Gordon Latta, ([Pólya and Latta, 1974](#)), and it is the subject of VCA’s Chapter 11: At each point z in $\mathbb{C}$ we draw a vector $P(z)$ that is defined to be the *complex conjugate* of $f(z)$, so $P(z) \equiv \overline{f(z)}$. This *Pólya vector field* then provides a wonderfully vivid means of *visualizing* the real and imaginary parts of the integral²⁴ of $f(z)$ along a directed path (aka a *contour*) C connecting two points in $\mathbb{C}$, namely, (11.1) on page 549:

$$\int_C f(z) dz = [\text{flow of } P \text{ along } C] + i [\text{flow of } P \text{ across } C].$$

Here, the imaginary flux component is *positive* if P flows across C *from our left to our right* as we face forward in the direction of travel along C .

Next, it follows immediately from the *Cauchy–Riemann equations*, (4.7), that there is a wonderfully vivid *physical interpretation* of the existence of the complex derivative (what I call the *amplitwist*), namely,

If and only if the amplitwist $f'(z)$ exists, then the Pólya vector field $P(z) \equiv \overline{f(z)}$ is divergence-free and curl-free.

This in turn immediately provides a marvellously *physical explanation* of one the central results of Complex Analysis, namely, *Cauchy’s Theorem*, (8.20), the simplest version of which states that

If the amplitwist f' exists everywhere on and inside a simple closed loop L , then $\oint_L f(z) dz = 0$.

But if L contains a singularity of $f(z)$, then the integral need not vanish. An example of central importance is $f(z) = (1/z)$, for which the Pólya vector field is the radial outward flow from a *source* at the origin. Choosing L to be an origin-centred circle

²² In fact there are many other innovations scattered across the work, most notably my introduction of a concept I christened the *complex curvature*, which I apply to central force fields in Chapter 5, and to the *geometry* of harmonic functions in Chapter 12; there’s also the *Topological Argument Principle* in Chapter 7, and more besides, but I cannot attempt to catalogue *all* these ideas and discoveries here. Some of the observations in Chapter 12 were previously published in [Needham \(1994\)](#), which won the MAA’s *Carl B. Allendoerfer Award* in 1995.

²³ See [Braden \(1985\)](#), [Braden \(1987\)](#)—which won the MAA’s *Carl B. Allendoerfer Award* in 1988—and [Braden \(1991\)](#).

²⁴ Rest assured that the following ideas will all be explained *ab initio* in the main text; this section of the Preface is addressed to experienced readers who are *already familiar with* the fundamentals of standard Complex Analysis.

K of radius r , traced counterclockwise, we obtain an *immediate visual and physical explanation* of the iconic fact that

$$\oint_K \frac{1}{z} dz = 2\pi i.$$

For clearly there is no flow along K , and since $P(z) = (1/\bar{z})$ flows orthogonally across K (from left to right) with speed $(1/r)$, its flux across K is $(2\pi r)(1/r) = 2\pi$. Furthermore, this physical interpretation explains why the value of the integral will *remain* $2\pi i$ if K is continuously deformed into a general loop encircling the source at the origin, so long as K does not *cross* that point as it is deformed.

Lastly, the ability of modern computers to quickly and easily draw the Pólya vector field of any explicit formula for $f(z)$ makes the concept all the more powerful as a means of *visualizing* Complex Analysis.

While the evangelical work of VCA (and of Braden's earlier papers) failed to have its desired effect in the 20th century, my fervent hope now is that by explicitly singling out this concept for praise in this new Preface, more people may take notice of it, and the gospel of the Pólya vector field may thereby be spread amongst multitudes of new believers in the 21st century!

The Quiet Revolution

In hindsight, something was clearly in the air. No sooner had VCA been published than I began to notice the emergence of a wave of kindred works that likewise challenged the prevailing dominance of an arid, purely formal approach to mathematics, and that instead embraced intuitive explanations, the meaning of results, and, crucially, the revelatory power of *geometry*.

It appeared that I had unwittingly been enlisted into a resistance movement of sorts, one in which cell members were not permitted to know each others identities—perhaps for their own safety?! But, unlike Bourbaki, this quiet revolution had no name and no leaders; it was, to coin a phrase, “By the people, for the people.” Long may this healthy embrace of *meaning, intuition, and geometry* continue!

Cheers!

I raise my glass to the *next* 25 years!

T. N.
Mill Valley, California
June, 2022

PREFACE

Theories of the known, which are described by different physical ideas, may be equivalent in all their predictions and hence scientifically indistinguishable. However, they are not psychologically identical when trying to move from that base into the unknown. For different views suggest different kinds of modifications which might be made and hence are not equivalent in the hypotheses one generates from them in one's attempt to understand what is not yet understood.

Feynman (1966)

A Parable

Imagine a society in which the citizens are encouraged, indeed compelled up to a certain age, to read (and sometimes write) musical scores. All quite admirable. However, this society also has a very curious—few remember how it all started—and disturbing law: *Music must never be listened to or performed!*

Though its importance is universally acknowledged, for some reason music is not widely appreciated in this society. To be sure, professors still excitedly pore over the great works of Bach, Wagner, and the rest, and they do their utmost to communicate to their students the beautiful meaning of what they find there, but they still become tongue-tied when brashly asked the question, “What’s the point of all this?!”

In this parable, it was patently unfair and irrational to have a law forbidding would-be music students from experiencing and understanding the subject directly through “sonic intuition.” But in our society of mathematicians we *have* such a law. It is not a written law, and those who flout it may yet prosper, but it says, *Mathematics must not be visualized!*

More likely than not, when one opens a random modern mathematics text on a random subject, one is confronted by abstract symbolic reasoning that is divorced from one’s sensory experience of the world, *despite* the fact that the very phenomena one is studying were often discovered by appealing to geometric (and perhaps physical) intuition.

This reflects the fact that steadily over the last hundred years the honour of visual reasoning in mathematics has been besmirched. Although the great mathematicians have always been oblivious to such fashions, it is only recently that the “mathematician in the street” has picked up the gauntlet on behalf of geometry.

The present book openly challenges the current dominance of purely symbolic logical reasoning by using new, visually accessible arguments to explain the truths of elementary complex analysis.

Computers

In part, the resurgence of interest in geometry can be traced to the mass-availability of computers to draw mathematical objects, and perhaps also to the related, somewhat breathless, popular interest in chaos theory and in fractals. This book instead advocates the more sober use of computers as an aid to geometric *reasoning*.

I have tried to encourage the reader to think of the computer as a physicist would his laboratory—it may be used to check existing ideas about the construction of the world, or as a tool for discovering new phenomena which then demand new ideas for their explanation. Throughout the text I have suggested such uses of the computer, but I have deliberately avoided giving *detailed* instructions. The reason is simple: whereas a mathematical idea is a timeless thing, few things are more ephemeral than computer hardware and software.

Having said this, the program “ $f(z)$ ” is currently²⁵ the best tool for visually exploring the ideas in this book; a free demonstration version can be downloaded directly from Lascaux Graphics. On occasion it would also be helpful if one had access to an all-purpose mathematical engine such as *Maple* or *Mathematica*. However, I would like to stress that none of the above software is essential: the entire book can be fully understood without *any* use of a computer.

Finally, some readers may be interested in knowing how computers were used to produce this book. Perhaps five of the 501 diagrams were drawn using output from *Mathematica*; the remainder I drew by hand (or rather “by mouse”) using CorelDRAW, occasionally guided by output from “ $f(z)$ ”. I typeset the book in $\text{\LaTeX}$ using the wonderful *Y&Y T_EX System for Windows*²⁶, the figures being included as EPS files. The text is Times²⁷, with Helvetica heads, and the mathematics is principally MathTime, though nine other mathematical fonts make cameo appearances. All of these Adobe Type 1 fonts were obtained from Y&Y, Inc., with the exception of Adobe’s *MathematicalPi-Six* font, which I used to represent quaternions. Finally, OUP printed the book directly from my PostScript file.

The Book’s Newtonian Genesis

It is fairly well known that Newton’s original 1665 version of the calculus was different from the one we learn today: its essence was the manipulation of power

²⁵ Sadly, this no longer exists.

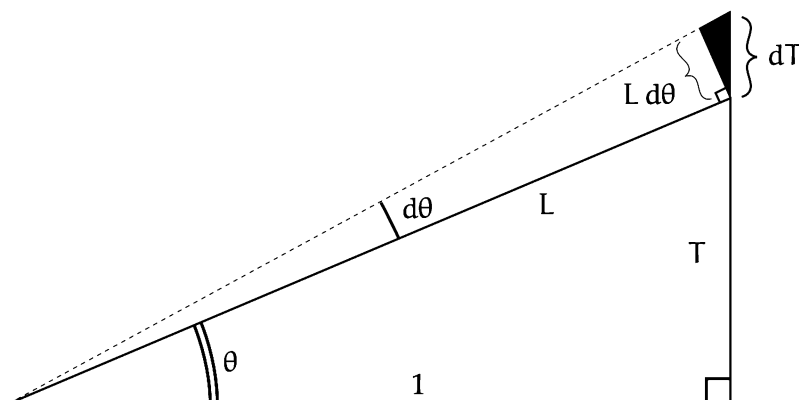
²⁶ Sadly, this no longer exists.

²⁷ In this new edition, the headings are Optima, the text is Palatino, and the mathematics is Euler.

series, which Newton likened to the manipulation of decimal expansions in arithmetic. The symbolic calculus—the one in every standard textbook, and the one now associated with the name of Leibniz—was also perfectly familiar to Newton, but apparently it was of only incidental interest to him. After all, armed with his power series, Newton could evaluate an integral like $\int e^{-x^2} dx$ just as easily as $\int \sin x dx$. Let Leibniz try *that*!

It is less well known that around 1680 Newton became disenchanted with both these approaches, whereupon he proceeded to develop a *third* version of calculus, based on *geometry*. This “geometric calculus” is the mathematical engine that propels the brilliant physics of Newton’s *Principia*.

Having grasped Newton’s method, I immediately tried my own hand at using it to simplify my teaching of introductory calculus. An example will help to explain what I mean by this. Let us show that if $T = \tan \theta$, then $\frac{dT}{d\theta} = 1 + T^2$. If we increase θ by a small amount $d\theta$ then T will increase by the amount dT in the figure below. To obtain the result, we need only observe that in the limit as $d\theta$ tends to zero, the black triangle is ultimately similar [exercise] to the shaded triangle. Thus, in this limit,



$$\frac{dT}{L d\theta} = \frac{L}{1} \implies \frac{dT}{d\theta} = L^2 = 1 + T^2.$$

Only gradually did I come to realize how naturally this mode of thought could be applied—almost exactly 300 years later!—to the geometry of the complex plane.

Reading This Book

In the hope of making the book fun to read, I have attempted to write as though I were explaining the ideas directly to a friend. Correspondingly, I have tried to make you, the reader, into an active participant in developing the ideas. For example, as an argument progresses, I have frequently and deliberately placed a pair of logical stepping stones sufficiently far apart that you may need to pause

and stretch slightly to pass from one to the next. Such places are marked “[exercise]”; they often require nothing more than a simple calculation or a moment of reflection.

This brings me to the exercises proper, which may be found at the end of each chapter. In the belief that the essential prerequisite for finding the answer to a question is the *desire* to find it, I have made every effort to provide exercises that provoke curiosity. They are considerably more wide-ranging than is common, and they often establish important facts which are then used freely in the text itself. While problems whose be all and end all is routine calculation are thereby avoided, I believe that readers will automatically develop considerable computational skill *in the process* of seeking solutions to these problems. On the other hand, my intention in a large number of the exercises is to illustrate how geometric thinking can often *replace* lengthy calculation.

Any part of the book marked with a star (“*”) may be omitted on a first reading. If you do elect to read a starred section, you may in turn choose to omit any starred *subsections*. Please note, however, that a part of the book that is starred is not necessarily any more difficult, nor any less interesting or important, than any other part of the book.

Teaching from this Book

The entire book can probably be covered in a year, but in a single semester course one must first decide what *kind* of course to teach, then choose a corresponding path through the book. Here I offer just three such possible paths:

- **Traditional Course.** Chapters 1 to 9, *omitting all starred material* (e.g., the whole of Chapter 6).
- **Vector Field Course.** In order to take advantage of the Pólya vector field approach to visualizing complex integrals, one could follow the “Traditional Course” above, omitting Chapter 9, and adding the unstarred parts of Chapters 10 and 11.
- **Non-Euclidean Course.** At the expense of teaching any integration, one could give a course focused on Möbius transformations and non-Euclidean geometry. These two related parts of complex analysis are probably the most important ones for contemporary mathematics and physics, and yet they are also the ones that are almost entirely neglected in undergraduate-level texts. On the other hand, graduate-level works tend to assume that you have already encountered the main ideas as an undergraduate: Catch 22!

Such a course might go as follows: All of Chapter 1; the unstarred parts of Chapter 2; all of Chapter 3, including the starred sections but (possibly) omitting

the starred *subsections*; all of Chapter 4; all of Chapter 6, including the starred sections but (possibly) omitting the starred *subsections*.

Omissions and Apologies

If one believes in the ultimate unity of mathematics and physics, as I do, then a very strong case for the necessity of complex numbers can be built on their apparently fundamental role in the quantum mechanical laws governing *matter*. Also, the work of Sir Roger Penrose has shown (with increasing force) that complex numbers play an equally central role in the relativistic laws governing the structure of *space-time*. Indeed, if the laws of matter and of spacetime are ever to be reconciled, then it seems very likely that it will be through the auspices of the complex numbers. This book cannot explore these matters; instead, we refer the interested reader to [Feynman \(1963, Vol. III\)](#), [Feynman \(1985\)](#), [Penrose and Rindler \(1984\)](#), and especially [Penrose \(2005\)](#).

A more serious omission is the lack of discussion of *Riemann surfaces*, which I had originally intended to treat in a final chapter. This plan was aborted once it became clear that a serious treatment would entail expanding the book beyond reason. By this time, however, I had already erected much of the necessary scaffolding, and this material remains in the finished book. In particular, I hope that the interested reader will find the last three chapters helpful in understanding Riemann's original physical insights, as expounded by [Klein \(1881\)](#). See also [Springer \(1981, Ch. 1\)](#), which essentially reproduces Klein's monograph, but with additional helpful commentary.

I consider the history of mathematics to be a vital tool in understanding both the current state of mathematics, and its trajectory into the future. Sadly, however, I can do no more than touch on historical matters in the present work; instead I refer you to the remarkable book, *Mathematics and Its History*, by John [Stillwell \(2010\)](#). Indeed, I strongly encourage you to think of his book as a companion to mine: not only does it trace and explain the development of complex analysis, but it also explores and illuminates the connections with other areas of mathematics.

To the expert reader I would like to apologize for having invented the word "amplitwist" [Chapter 4] as a synonym (more or less) for "derivative", as well the component terms "amplification" and "twist". I can only say that the need for *some* such terminology was forced on me in the classroom: if you try teaching the ideas in this book *without* using such language, I think you will quickly discover what I mean! Incidentally, a precedence argument in defence of "amplitwist" might be that a similar term was coined by the older German school of Klein, Bieberbach, *et al.* They spoke of "eine Drehstreckung", from "drehen" (to twist) and "strecken" (to stretch).

A significant proportion of the geometric observations and arguments contained in this book are, to the best of my knowledge, new. I have not drawn attention to this in the text itself as this would have served no useful purpose: students don't need to know, and experts will know without being told. However, in cases where an idea is clearly unusual but I am aware of it having been published by someone else, I have tried to give credit where credit is due.

In attempting to rethink so much classical mathematics, I have no doubt made mistakes; the blame for these is mine alone. Corrections will be gratefully received at VCA.correction@gmail.com.

My book will no doubt be flawed in many ways of which I am not yet aware, but there is one "sin" that I have intentionally committed, and for which I shall not repent: many of the arguments are not rigorous, at least as they stand. This is a serious crime if one believes that our mathematical theories are merely elaborate mental constructs, precariously hoisted aloft. Then rigour becomes the nerve-racking balancing act that prevents the entire structure from crashing down around us. But suppose one believes, as I do, that our mathematical theories are attempting to capture aspects of a robust Platonic world that is not of our making. I would then contend that an initial lack of rigour is a small price to pay if it allows the reader to see into this world more directly and pleasurable than would otherwise be possible.

T. N.
San Francisco, California
June, 1996

ACKNOWLEDGEMENTS

First and foremost I wish to express my indebtedness to Dr. Stanley Nel. He is my friend, my colleague, and my Dean, and in all three of these capacities he has helped me to complete this book. As a friend he offered support when progress was slow and my spirits were low; as a mathematical colleague he read much of the book and offered helpful criticisms; as Dean he granted me a succession of increasingly powerful computers, and when the US Immigration Service sought to have my position filled by an “equally qualified” American, he successfully fought them on my behalf. For all this, and much else besides, I offer him my deep gratitude.

Next I would like to thank Prof. John Stillwell of Monash University. The great value I place on his writings should be clear from the frequency with which I refer to them in the pages that follow. Also, though I lack his gift for conciseness, I have sought to emulate elements of his approach in an attempt to give back *meaning* to mathematical concepts. Finally, my greatest and most concrete debt arises from the fact that he read each draft chapter as it was written, and this despite the fact that we had never even met! The book owes a great deal to his numerous helpful suggestions and corrections.

I consider myself very fortunate that the mathematics department here at the University of San Francisco is completely free of political intrigue, rivalry, and other assorted academic blights. I am grateful to *all* my colleagues for creating such a friendly and supportive atmosphere in which to work. In particular, however, I should like to single out the following people for thanks:

- Nancy Campagna for her diligent proof-reading of half the book;
- Allan Cruse and Millianne Lehmann, not only for granting all my software requests during their respective tenures as Department Chair, but also for all their kind and sage advice since my arrival in the United States;
- James Finch for his patience and expertise in helping me overcome various problems associated with my typesetting of the book in $\text{\LaTeX}$;
- Robert Wolf for having built up a superb mathematics collection in our library;
- Paul Zeitz for his great faith in me and in the value of what I was trying to accomplish, for his concrete suggestions and corrections, and for his courage in being the first person (other than myself) to teach complex analysis using chapters of the book.

Prof. Gerald Alexanderson of Santa Clara University has my sincere thanks for the encouragement he offered me upon reading some of the earliest chapters, as well as for his many subsequent acts of kindness.

I will always be grateful for the education I received at Merton College, Oxford. It is therefore especially pleasing and fitting to have this book published by OUP, and I would particularly like to thank Dr. Martin Gilchrist, the former Senior Mathematics Editor, for his enthusiastic encouragement when I first approached him with the idea of the book.

When I first arrived at USF from England in 1989 I had barely seen a computer. The fact that OUP printed this book directly from my Internet-transmitted PostScript files is an indication of how far I have come since then. I owe all this to James Kabage. A mere graduate student at the time we met, Jim quickly rose through the ranks to become Director of Network Services. Despite this fact, he never hesitated to spend *hours* with me in my office resolving my latest hardware or software crisis. He always took the extra time to clearly explain to me the reasoning leading to his solution, and in this way I became his student.

I also thank Dr. Benjamin Baab, the Executive Director of Information Technology Services at USF. Despite his lofty position, he too was always willing to roll up his sleeves in order to help me resolve my latest Microsoft conundrum.

Eric Scheide (our multitalented Webmaster) has my sincere gratitude for writing an extremely nifty *Perl* program that greatly speeded my creation of the index.

I thank Prof. Berthold Horn of MIT for creating the magnificent *Y&Y T_EX System for Windows*²⁸, for his generous help with assorted T_EXnical problems, and for his willingness to adopt my few suggestions for improving what I consider to be the Mercedes-Benz of the T_EX world.

Similarly, I thank Martin Lapidus of Lascaux Graphics for incorporating many of my suggestions into his "*f(z)*"²⁹ program, thereby making it into an even better tool for doing "visual complex analysis".

This new printing of the book incorporates a great many corrections. Most of these were reported by readers, and I very much appreciate their efforts. While I cannot thank each one of these readers by name, I must acknowledge Dr. R. von Randow for single-handedly having reported more than 30 errors.

As a student of Roger Penrose I had the privilege of watching him think out loud by means of his beautiful blackboard drawings. In the process, I became convinced that if only one tried hard enough—or were clever enough!—every mathematical mystery could be resolved through geometric reasoning. George Burnett-Stuart and I became firm friends while students of Penrose. In the course of our endless discussions of music, physics, and mathematics, George helped me to refine both

²⁸ While *Y & Y* no longer exists, my gratitude lives on.

²⁹ While "*f(z)*" no longer exists, my gratitude lives on.

my conception of the nature of mathematics, and of what constitutes an acceptable explanation within that subject. My dedication of this book to these two friends scarcely repays the great debt I owe them.

The care of several friends helped me to cope with depression following the death of my beloved mother Claudia. In addition to my brother Guy and my father Rodney, I wish to express my appreciation to Peter and Ginny Pacheco, and to Amy Miller. I don't know what I would have done without their healing affection.

Lastly, I thank my dearest wife Mary. During the writing of this book she allowed me to pretend that science was the most important thing in life; now that the book is over, she is my daily proof that there is something even more important.

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